

A Appendix: Extended Definitions of Systems Concepts

System: A system is a group of interacting or interrelated entities that form a unified whole. A system is delineated by its spatial and temporal boundaries, surrounded and influenced by its environment, described by its structure and purpose, and expressed through its functioning. Systems are the subjects of study of systems theory.

Complex System: A complex system is a system where the behavior of the whole within a given environment cannot be understood directly by understanding the behavior of the parts and the environment as separate additive entities. Rather the system level behavior is deeply dependant on the relations between the parts. Complex systems are systems whose behavior is intrinsically difficult to model due to the dependencies, competitions, relationships, or other types of interactions between their parts or between a given system and its environment. Systems that are "complex" have distinct properties that arise from these relationships, such as non-linearity, emergence, spontaneous order, adaptation, and feedback loops, among others. In practice, complex systems exhibit unintuitive behaviors when acted upon. In many cases it is useful to represent such a system as a model (for example a network where the nodes represent the components and the links their interactions). Because such systems appear in a wide variety of fields, the commonalities among them have become the topic of their own independent area of research.

Model: A model is a graphical, mathematical (symbolic), physical, verbal representation, or simplified version of a concept, phenomenon, relationship, structure, system, or an aspect of the real world. The objectives of a model include (1) to facilitate understanding by eliminating unnecessary components, (2) to aid in decision making by simulating 'what if' scenarios, (3) to explain, control, and predict events on the basis of past observations. Since most objects and phenomenon are very complicated (have numerous parts) and are much too complex (parts have dense interconnections) to be understood in their entirety, a model contains only those features that are of primary importance to the model maker's purpose.

Formal model: A formal model is a model in which the description is complete and non-ambiguous. It has well-formed syntax and semantics, such that it is amenable to systematic (usually automatable) processing and analysis subject to logical rules. It is based on rigorous methods and formats; it often enables analysis to be performed on the models.

Mathematical model: A mathematical model is a special type of formal model, and a description of a system using mathematical concepts and language.

Computational model: A computational model is a special type of a mathematical model which is expressed in a programming language. It allows one to run simulations, or backtest against data.

A Digital Twin: A computational model integrated with data from an implemented system that aids the operations and governance of that implemented system. Digital twins support both automated operations and human governing mechanisms, increasing visibility into the systemic effects of using specific instruments to change the system's behavior.

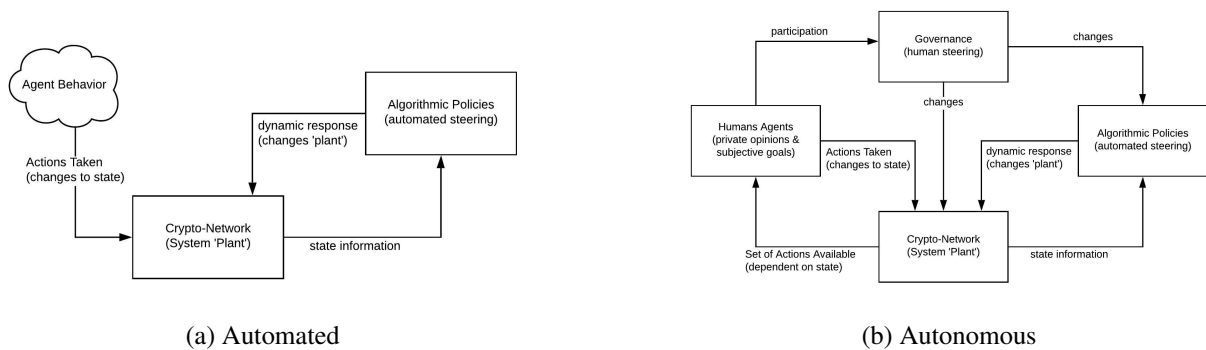


Figure 15: Automated versus Autonomous systems.

Automated System: A closed feedback loop between a computational system and another system being controlled that dynamically enforces a set of rules with the intent to drive a particular system behavior. The goals and behavior of agents are extrinsic to the automated system. Decentralized network software provide an automated system.

Autonomous System: A closed feedback loop at a sufficiently high level of abstraction as to require no additional inputs in order to act. The goals and behaviors of an autonomous system are intrinsic.

Economic System: An autonomous system including but not limited to assets, mechanisms, agents, instruments, and policy makers. An economy may include automated software as part of its infrastructure, but the economic system is characterized by the interplay between the human and machine elements.

Commentary Regarding Models:

- Computational models are used extensively to explore complex systems because complex systems exhibit both non-ergodicity and non-linearity. Both non-ergodicity and non-linearity limit our capacity to extrapolate our past observations about a system's past behavior into predictions about future behavior. Understanding the boundaries of our models is paramount to using them properly.
- Models are subjective framing and modeling choices that directly effect the conclusions one reaches when using them. This is true of all fields and places an onus on the systems engineer to make choices which improve the usefulness of the models, even at the cost of precision. Models should be as simple as possible but no simpler.
- While there is no optimal model *per se*, it is not fair to say all models are equal. In practice, models are judged by *usefulness* where usefulness is measured relative to some desired end. A model is useful when its application in the manner described above facilitates the operation and governance of the economic system in accordance with its goals.

B Appendix: On Controls and Policies in Economies

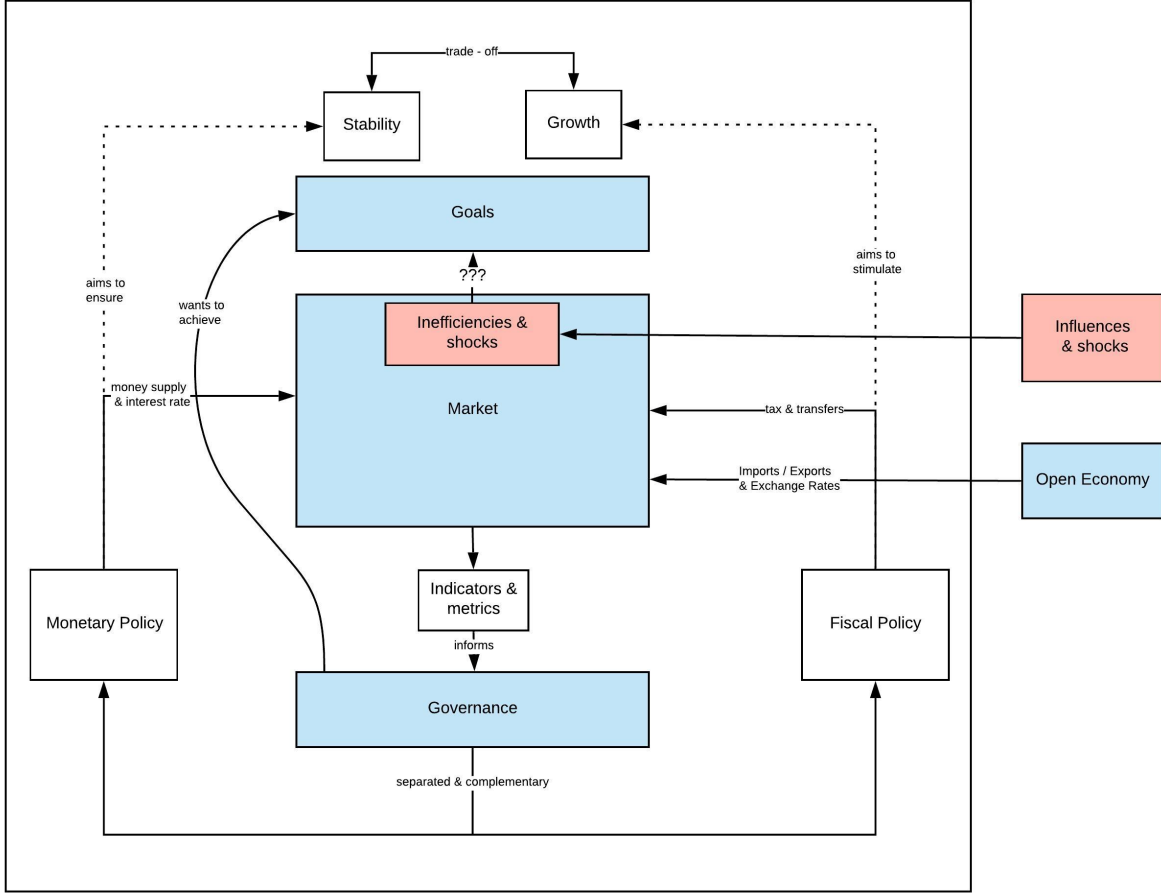


Figure 16: Government impact markets with the aim of guidance towards its set objectives.

An economy is a rule-based trade environment in which humans organize their interactions to pursue common goals. These goals are typically set by the society in a democratic process, and compliance with the targets is monitored by the directly elected governance entity. How well the economy performs with respect to its goals is tracked by economic indicators and health metrics as described above. In order to steer the society towards its goals, if the metrics suggest otherwise, policy instruments exist which may be used to intervene in the markets. These instruments are used to enable corrections caused by market inefficiencies, as well as internal and external shocks to the economy. (Figure 17) demonstrates the process of governance intervention on markets with the aim of guidance towards the set objectives.

In the most abstract sense, the economy desires to achieve the state of stability and growth. However, trade-offs between these goals exist, since both cannot be reached at the same time to a full extent because measures necessary to get there partially contradict each other. To ensure the appropriate balance between both forces, in most evolved economies the mechanisms driving growth and stability are subject to supervision of two distinct entities: the *monetary authority* and the *fiscal authority*. They engage in a coordination game and base their decisions on their counterpart's actions as well as the conditions of the economy.

The monetary authority, most often a central bank or the currency board, sets a monetary policy by choosing parameters for controlling money supply and the short-term interest rate within the economy. Their main goal is to ensure price stability relative to factors both inside and outside of the economy, observing the changes in prices of consumption goods and the foreign exchange rate.

The fiscal authority, most often a representative-based government, sets a fiscal policy to ensure stimulation of growth and redistribution of wealth in accordance with the economic goals. To do so, it sets taxes and provides social transfers within the economy and decides the amount of government spending. A fiscally responsible government will make sure not to go into debt and thus place a burden on future generations.

These interventions, but also simple market inefficiencies and shocks, make the economy react in business cycles of alternating phases of growth and recession.

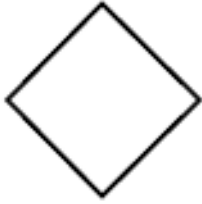
C Appendix: Economic System Specification Syntax

Setting the boundaries of a system defines which of the component processes are considered internal and which are external. External processes may affect a defined system but should not directly interact with the internal system processes. The systems view specifies a set of intellectual tools that can be applied to any sized set of processes, resources, and information. The systems approach simplifies problems, describes for significant processes, and identifies controls. The system view is used to create simulations of future conditions. Another of the benefits of this approach is that it clearly identifies the assumptions on which simulations are based. A systems model is both a valuable research tool and a platform for communication and decision-making. Thus, a carefully constructed system model is necessary for capturing the structural dynamics of an economic system.

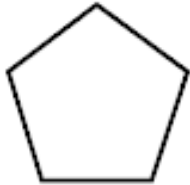
Colors in the Economic System Specification refer to:

	Flow of DOTs
	Interface between Subsystems
	Economic Subsystem
	External Process
	Uncontrolled Behavior
	Controlled or Automated Behavior
	State Information

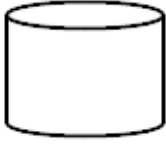
Symbols in the Economic System Specification refer to:



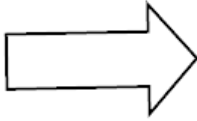
Mechanism Action: A method exposed to a set of actors, which can include agents, expected populations, and controlling entities. The mechanism receives user actions and input, employing state information and control policy parameters to provide an update to a defined set of state variables.



Mechanism Control: The action space available exclusively to the controlling entity. May be actively controlled through a strategy developed using information gathered from available publicly and privately known state variables, metrics, external factors, internal beliefs, and utility functions. The control mechanism may change controllable parameters and/or invoke a phase change of the system via the control.



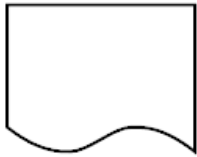
State Variable: Defined to capture the shape and property of the network, representing publicly or privately held output of the system. Mechanisms provide updates to the state variable. The set of all state variables defines the state of the system.



Conserved Flow: A flow represents the mass of currency movement. The currency must be the same as the stock at either end. The link identifies the relationship between nodes. The flow is conserved through accounting for all sources and sinks of the currency.



Information Flow: Any data structure used to convey a decision to a mechanism action or a control mechanism. The information may be the result of a change from external processes, a state update, or a control policy change.



Decision/Process: Logic that informs action on an action mechanism, control mechanism, or other decision. The condition accepts inputs from a subset of state variables, metrics, external factors, internal beliefs, policy goals, and utility functions.



External Perspective: Information from the outside world that affects user action control policy decisions on the internal system. This feed may be based purely on external factors or may represent the external view of the performance of the internal system.



Detailed Relation: External dynamic not part of the internal system. However, it may contain a level of complexity and exert critical influence on the internal system to require further development.

D Appendix: Agent-Role Data Models

For economic systems comprised of multiple mechanisms, agents will have more advanced strategies including the ability to manipulate those mechanisms through mixed strategies that use one mechanism to create exploitative opportunities over other mechanisms. In this framework, an agent is mapped to a role if that agent has a strategy relating that role. It may be that the conditions for an action with respect to the role are never met, but as long as there is a condition under which the agent would take such an action there is a mapping between the agent and the role.

In order to build a model capable of exploring multi-mechanism strategies it is necessary to maintain information about each agents strategy across each role. The multi-graph model contains nodes and edges. Nodes have two types: (1) Agent and (2) Role. Edges generally have two types: (1) Agent-Agent, (2) Agent-Role, though in principal Role-Role edges are also well defined as needed. There are also more advanced relations such as agent-agent-role which are required to manage data such as stakes in systems where agents can stake to delegate authority to another agent.

These multi-graph models allow us to enforce token conservation laws, encode complex multi-mechanism strategies, view the state from an agent-centric view or from a role-centric view while maintaining internal consistency. See Figure 17 for a visual overview of the multi-graph agent-role data model.

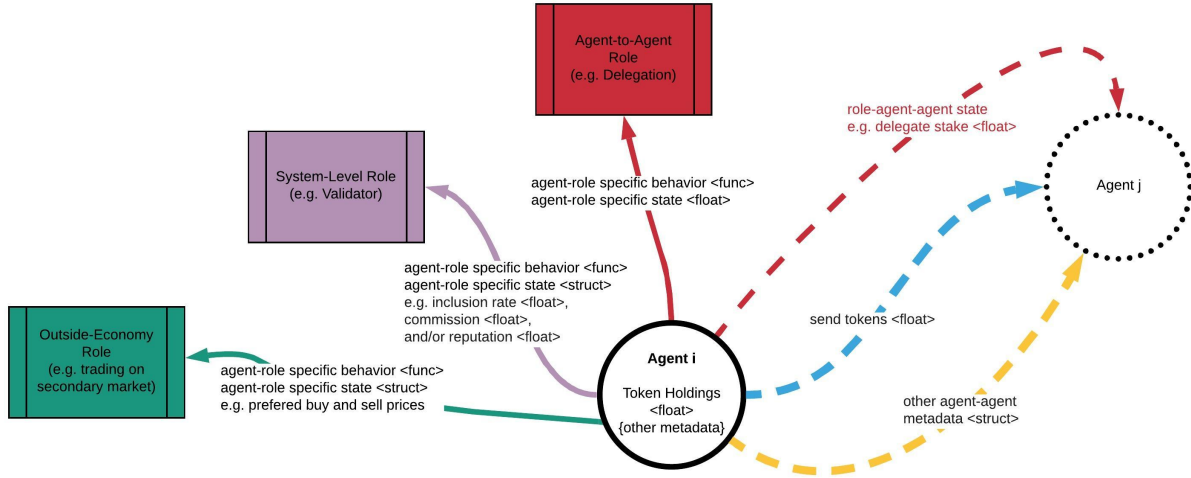


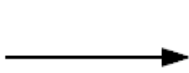
Figure 17: The multi-graph data model has two types of nodes, agents and roles. Edges encode relationships between agents and agents, as well as agents and roles.

E Appendix: Differential Specification Syntax

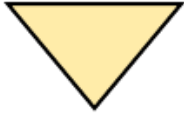
The differential specification is effectively described through a system modelling syntax. The symbols referenced in the differential specification diagram are described here. Note that all information flows described are feed-forward, feedback of state to any process is implicit. Furthermore, the differential specification is laid on a grid where the rows are:

- The **Policies** determine the inputs to the system dynamics regardless of whether they come from the outside environment, user behavior or algorithmic decision systems.
- The **State Update Logic** determines how the state of the system changes as a result of the inputs provided. Note that not all outcomes are the outcomes intended by the actor (system or agent) when the input decision was made due to the interdependence of these inputs.
- The **State** is flattened and summarized in order to keep track of the the interdependence of the mechanisms on the same state variables. Note that the full data model for the state space is generally a network object for which the shown states are mid-level aggregates or summary statistics.
- The **Metrics** are values computed from the state variables in order to assess the economic system. In a computational model, metrics may be used for governance decisions, algorithmic policies or to inform the behavior of individual agents.

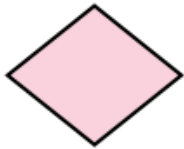
The columns in a differential specification refer to the **Partial State Update Block** also called *substeps*, which break the logic up into an ordered sequence in order to support second order dependencies. For example, there may be a set of user behavior functions that resolve a decision to participate and incur some costs, and then a set of environmental process drivers that determine the rewards those participants receive as a result of all of their decisions and the events which transpire there after. A simple differential specification will have one or two substeps, whereas a complex economic system could have 8 to 10 substeps.



Information Flow: Any data structure may be used to convey state update execution. The expected path is from actions, policies, and processes to mechanisms to states variables to metrics.



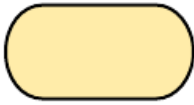
Environmental Process Driver: Data feed that informs an exogenous process. This may be real or assumed data to provide updated input for a model representing a change of external conditions.



User Behavior Function: A population or agent-based strategy developed using information gathered from available state variables, metrics, external factors, internal beliefs, and utility functions. The action space available to achieve the strategy is executed through the mechanisms. No action may be fulfillment of the strategy. The user behavior/action should not directly be a state variable.



Control Function: The controllable action space under the controlling entity. May be actively controlled through a strategy developed using information gathered from available publicly and privately known state variables, metrics, external factors, internal beliefs, and utility functions. The action space available to achieve the strategy is executed through the mechanisms. Passive control (no action) may correctly model this behavior. Policy functions should not directly modify a state variable, but may affect any parameters in a mechanism as well as the availability of a mechanism.



Environmental Process Model: A modelled generator providing an update to state variables using external input signals from an environmental process driver. User action behavior and control functions should not affect this model. The affected state variables should be purely exogenously based, but may be blended from the internal state if necessary to model the system. The exogenous state variables are the link between the changes from the outside world germane to the modeled system, and provides the information necessary for users and controllers to make decisions.



Mechanism: System defined functions that take user actions and control policy input to update a set of state variables. Mechanism functions may be logically or operationally grouped to be executed concurrently as a partial state update.



State Variable: Defined to capture the shape and property of the network, representing publicly or privately held output of the system. The updates to the information can only come through mechanisms and exogenous models and not user or control actions. The data may be updated repeatedly within a simulated step. The resolution of all partial updates is the state update.



Performance Indicator: Metrics calculated from a subset of state variables. The indicator may represent actual publicly or privately available data. The metric could also be a computed signal from the state used to determine the condition of a control policy. Also, this may be an expected indicator employed by users to execute on their strategy. Care must be taken to update the metric after the input state variable(s) are updated as to obtain the correct result within a state update.

References

- [1] Dr. Gavin Wood. Polkadot: Vision for a heterogeneous multi-chain framework. <https://polkadot.network/PolkaDotPaper.pdf>, 2019.
- [2] Iman Jahandideh, Fatemeh Ghassemi, and Marjan Sirjani. Hybrid rebeca: Modeling and analyzing of cyber-physical systems. In Roger Chamberlain, Walid Taha, and Martin Törngren, editors, *Cyber Physical Systems. Model-Based Design*, pages 3–27, Cham, 2019. Springer International Publishing.
- [3] Marten Lohstroh, Patricia Derler, and Marjan Sirjani. *Principles of Modeling*, volume 10760. Springer: Lecture Notes in Computer Science book series, 2018.
- [4] John Sterman. Business dynamics, system thinking and modeling for a complex world. [http://lst-iiep.iiep-unesco.org/cgi-bin/wwwi32.exe/\[in=epidoc1.in\]/?t2000=013598/\(100\)](http://lst-iiep.iiep-unesco.org/cgi-bin/wwwi32.exe/[in=epidoc1.in]/?t2000=013598/(100), 19, 01 2000), 19, 01 2000.
- [5] Michael Zargham, Zixuan Zhang, and Victor Preciado. A state-space modeling framework for engineering blockchain-enabled economic systems. In *Proceeding of the International Complex Systems Conference*. New England Complex Systems Institute, 2018.
- [6] Z. Zhang, M. Zargham, and V.M. Preciado. On modeling blockchain-enabled economic networks as stochastic dynamical systems. *Applied Network Science: Blockchain and Cryptocurrency [accepted]*, 2019 [forthcoming].
- [7] Jamsheed Shorish. Blockchain state machine representation. SocArXiv, Janu 2018.
- [8] Dmitry Novikov. *Systems Theory and Systems Analysis. Systems Engineering*, volume 47, pages 39–44. 01 2016.
- [9] S. Voshmgir and M. Zargham. Cryptoeconomics research roadmap: A cybernetics approach to token economics and economic systems engineering. Vienna University of Economics and Business Working Paper Series, 2019.
- [10] A Wayne Wymore. *Model-based systems engineering*. CRC press, 1993.
- [11] Yaneer Bar-Yam. When systems engineering fails-toward complex systems engineering. In *SMC'03 Conference Proceedings. 2003 IEEE International Conference on Systems, Man and Cybernetics. Conference Theme-System Security and Assurance (Cat. No. 03CH37483)*, volume 2, pages 2021–2028. IEEE, 2003.
- [12] Yaneer Bar-Yam. General features of complex systems. *Encyclopedia of Life Support Systems (EOLSS)*, UNESCO, EOLSS Publishers, Oxford, UK, 1, 2002.
- [13] Daniel DeLaurentis. Role of humans in complexity of a system-of-systems. In Vincent G. Duffy, editor, *Digital Human Modeling*, pages 363–371, Berlin, Heidelberg, 2007. Springer Berlin Heidelberg.
- [14] Iohan Vargas, Thiago Gottardi, and Rosana Braga. Approaches for integration in system of systems: a systematic review. pages 32–38, 05 2016.
- [15] Radhakisan Baheti and Helen Gill. Cyber-physical systems. *The impact of control technology*, 12(1):161–166, 2011.
- [16] James M. Tien. On integration and adaptation in complex service systems. *Journal of Systems Science and Systems Engineering*, 17(4):385–415, Dec 2008.
- [17] Lotfi A. Zadeh. Outline of a new approach to the analysis of complex systems and decision processes. *IEEE Trans. Systems, Man, and Cybernetics*, 3(1):28–44, 1973.
- [18] J. S. Shamma. Robust stability with time-varying structured uncertainty. *IEEE Transactions on Automatic Control*, 39(4):714–724, April 1994.
- [19] J.N. Tsitsiklis, D.P. Bertsekas, and M. Athans. Distributed asynchronous deterministic and stochastic gradient optimization algorithms. *IEEE Transactions on Automatic Control*, 31(9):803–812, 1986.
- [20] M. Chiang, S.H. Low, A.R. Calderbank, and J.C. Doyle. Layering as optimization decomposition: A mathematical theory of network architectures. *Proceedings of the IEEE*, 95(1):255–312, 2007.
- [21] Angelia Nedic, Asuman Ozdaglar, and Pablo A Parrilo. Constrained consensus and optimization in multi-agent networks. *IEEE Transactions on Automatic Control*, 55(4):922–938, 2010.
- [22] Reza Olfati-Saber and Richard M. Murray. Distributed cooperative control of multiple vehicle formations using structural potential functions. *IFAC Proceedings Volumes*, 35(1):495 – 500, 2002. 15th IFAC World Congress.
- [23] A. Jadbabaie, J. Lin, and S. Morse. Coordination of groups of mobile autonomous agents using nearest neighbor rules. *IEEE Transactions on Automatic Control*, 48(6):988–1001, 2003.
- [24] Reza Olfati-Saber. Flocking for multi-agent dynamic systems: Algorithms and theory. *CALIFORNIA INST OF TECH PASADENA CONTROL AND DYNAMICAL SYSTEMS*, 2004.

- [25] Engelbert J Dockner, Steffen Jorgensen, Ngo Van Long, and Gerhard Sorger. *Differential games in economics and management science*. Cambridge University Press, 2000.
- [26] Avner Friedman. *Differential games*. Courier Corporation, 2013.
- [27] Yu-Chi Ho. Differential games, dynamic optimization, and generalized control theory. *Journal of Optimization Theory and Applications*, 6(3):179–209, 1970.
- [28] Rufus Isaacs. *Differential games: a mathematical theory with applications to warfare and pursuit, control and optimization*. Courier Corporation, 1999.
- [29] Tamer Basar and Geert Jan Olsder. *Dynamic noncooperative game theory*, volume 23. Siam, 1999.
- [30] J. R. Marden, G. Arslan, and J. S. Shamma. Joint strategy fictitious play with inertia for potential games. *IEEE Transactions on Automatic Control*, 54(2):208–220, Feb 2009.
- [31] David WK Yeung and Leon A Petrosjan. *Cooperative stochastic differential games*. Springer Science & Business Media, 2006.
- [32] Yongcan Cao, Wenwu Yu, Wei Ren, and Guanrong Chen. An overview of recent progress in the study of distributed multi-agent coordination. *IEEE Transactions on Industrial informatics*, 9(1):427–438, 2012.
- [33] Robert Axelrod and William D Hamilton. The evolution of cooperation. *science*, 211(4489):1390–1396, 1981.
- [34] Reza Olfati-Saber, J. Alex Fax, and Richard M. Murray. Consensus and cooperation in networked multi-agent systems. *Proceedings of the IEEE*, 2007.
- [35] Ernst Fehr and Klaus M Schmidt. A theory of fairness, competition, and cooperation. *The quarterly journal of economics*, 114(3):817–868, 1999.
- [36] J. R. Marden, G. Arslan, and J. S. Shamma. Cooperative control and potential games. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, 39(6):1393–1407, Dec 2009.
- [37] David W Johnson and Roger T Johnson. *Cooperation and competition: Theory and research*. Interaction Book Company, 1989.
- [38] Yoad Lewenberg, Yoram Bachrach, Yonatan Sompolsky, Aviv Zohar, and Jeffrey S Rosenschein. Bitcoin mining pools: A cooperative game theoretic analysis. In *Proceedings of the 2015 International Conference on Autonomous Agents and Multiagent Systems*, pages 919–927. International Foundation for Autonomous Agents and Multiagent Systems, 2015.
- [39] Raymond A DeCarlo, Michael S Branicky, Stefan Pettersson, and Bengt Lennartson. Perspectives and results on the stability and stabilizability of hybrid systems. *Proceedings of the IEEE*, 88(7):1069–1082, 2000.
- [40] Daron Acemoglu, Vasco M. Carvalho, Asuman Ozdaglar, and Alireza Tahbaz-Salehi. The network origins of aggregate fluctuations. *Econometrica*, 80(5):1977–2016, 2012.
- [41] Dion Harmon, Marco Lagi, Marcus AM de Aguiar, David D Chinellato, Dan Braha, Irving R Epstein, and Yaneer Bar-Yam. Anticipating economic market crises using measures of collective panic. *PloS one*, 10(7):e0131871, 2015.
- [42] Daron Acemoglu, Asuman Ozdaglar, and Alireza Tahbaz-Salehi. Systemic risk and stability in financial networks. *American Economic Review*, 105(2):564–608, February 2015.
- [43] Yaneer Bar-Yam. Multiscale variety in complex systems. *Complexity*, 9(4):37–45, 2004.
- [44] Yaneer Bar-Yam. A mathematical theory of strong emergence using multiscale variety. *Complexity*, 9(6):15–24, 2004.
- [45] Dan Braha and Yaneer Bar-Yam. Topology of large-scale engineering problem-solving networks. *Physical Review E*, 69(1):016113, 2004.
- [46] Christian Wernz and Andrew Henry. Multi-level coordination and decision-making in service operations. *Service Science*, 1(4):270–283, 2009.
- [47] Christian Wernz and Abhijit Deshmukh. Multiscale decision-making: Bridging organizational scales in systems with distributed decision-makers. *European Journal of Operational Research*, 202(3):828 – 840, 2010.
- [48] Robert E. Lucas. Econometric policy evaluation: A critique. *Carnegie-Rochester Conference Series on Public Policy*, 1:19 – 46, 1976.
- [49] Kenneth J Arrow, Hollis B Chenery, Bagicha S Minhas, and Robert M Solow. Capital-labor substitution and economic efficiency. *The review of Economics and Statistics*, 43(3):225–250, 1961.
- [50] Ole Peters and Alexander Adamou. *Ergodicity Economics*. London Mathematical Laboratory, 2018.

- [51] John B Taylor, Michael Woodford, and Harald Uhlig. *Handbook of macroeconomics*. Elsevier, 1999.
- [52] B. M. Friedman and Frank Hahn, editors. *Handbook of Monetary Economics*, volume 1. Elsevier, 1 edition, 1990.
- [53] Nicholas Barberis and Richard Thaler. A survey of behavioral finance. *Handbook of the Economics of Finance*, 1:1053–1128, 2003.
- [54] Edwin A Locke. Toward a theory of task motivation and incentives. *Organizational behavior and human performance*, 3(2):157–189, 1968.
- [55] Amos Tversky and Daniel Kahneman. Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and uncertainty*, 5(4):297–323, 1992.
- [56] Daniel Kahneman, Stewart Paul Slovic, Paul Slovic, and Amos Tversky. *Judgment under uncertainty: Heuristics and biases*. Cambridge university press, 1982.
- [57] John H Kagel and Alvin E Roth. *The handbook of experimental economics*, volume 2. Princeton university press, 2016.
- [58] Michael S Branicky. Multiple lyapunov functions and other analysis tools for switched and hybrid systems. *IEEE Transactions on automatic control*, 43(4):475–482, 1998.
- [59] Georgios Piliouras, Evdokia Nikolova, and Jeff S Shamma. Risk sensitivity of price of anarchy under uncertainty. *ACM Transactions on Economics and Computation (TEAC)*, 5(1):5, 2016.
- [60] Alvin E Roth and Ido Erev. Learning in extensive-form games: Experimental data and simple dynamic models in the intermediate term. *Games and economic behavior*, 8(1):164–212, 1995.
- [61] Ali Jadbabaie, Pooya Molavi, Alvaro Sandroni, and Alireza Tahbaz-Salehi. Non-bayesian social learning. *Games and Economic Behavior*, 76(1):210 – 225, 2012.
- [62] Daron Acemoglu and Asuman Ozdaglar. Opinion dynamics and learning in social networks. *Dynamic Games and Applications*, 1(1):3–49, 2011.
- [63] Ido Erev and Alvin E Roth. Predicting how people play games: Reinforcement learning in experimental games with unique, mixed strategy equilibria. *American economic review*, pages 848–881, 1998.
- [64] Jianshu Chen and Ali H Sayed. Diffusion adaptation strategies for distributed optimization and learning over networks. *IEEE Transactions on Signal Processing*, 60(8):4289–4305, 2012.
- [65] Ilan Lobel and Evan Sadler. Information diffusion in networks through social learning. *Theoretical Economics*, 10(3):807–851, 2015.
- [66] Michael Grieves and John Vickers. Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems. In *Transdisciplinary perspectives on complex systems*, pages 85–113. Springer, 2017.
- [67] Flo Conway, Jim Siegelman, and Gerald L Alexanderson. Dark hero of the information age: In search of norbert wiener, the father of cybernetics. *The Mathematical Intelligencer*, 28(2):70–73, 2006.
- [68] K. W. Hipel, M. M. Jamshidi, J. M. Tien, and C. C. White III. The future of systems, man, and cybernetics: Application domains and research methods. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, 37(5):726–743, Sep. 2007.
- [69] Thomas Fischer and Christiane Herr. *An Introduction to Design Cybernetics*, pages 1–23. 07 2019.
- [70] Richard Thaler. Toward a positive theory of consumer choice. *Journal of Economic Behavior & Organization*, 1(1):39–60, 1980.
- [71] Elinor Ostrom. A behavioral approach to the rational choice theory of collective action: Presidential address, american political science association, 1997. *American political science review*, 92(1):1–22, 1998.
- [72] Heinz Von Foerster. Ethics and second-order cybernetics. In *Understanding understanding*, pages 287–304. Springer, 2003.
- [73] Hoda Heidari, Claudio Ferrari, Krishna Gummadi, and Andreas Krause. Fairness behind a veil of ignorance: A welfare analysis for automated decision making. In *Advances in Neural Information Processing Systems*, pages 1265–1276, 2018.
- [74] Kenneth Joseph Arrow. The economics of agency. Technical report, STANFORD UNIV CA INST FOR MATHEMATICAL STUDIES IN THE SOCIAL SCIENCES, 1984.
- [75] Martin Wollschlaeger, Thilo Sauter, and Juergen Jasperneite. The future of industrial communication: Automation networks in the era of the internet of things and industry 4.0. *IEEE industrial electronics magazine*, 11(1):17–27, 2017.

- [76] Alasdair Gilchrist. *Industry 4.0: the industrial internet of things*. Apress, 2016.
- [77] Charith Perera, Chi Harold Liu, Srimal Jayawardena, and Min Chen. A survey on internet of things from industrial market perspective. *IEEE Access*, 2:1660–1679, 2014.
- [78] Jay Lee, Behrad Bagheri, and Hung-An Kao. A cyber-physical systems architecture for industry 4.0-based manufacturing systems. *Manufacturing letters*, 3:18–23, 2015.
- [79] Janusz J Sikorski, Joy Haughton, and Markus Kraft. Blockchain technology in the chemical industry: machine-to-machine electricity market. In *Applied Energy*, volume 195, pages 234–246. Elsevier, 2017.
- [80] Ragunathan Rajkumar, Insup Lee, Lui Sha, and John Stankovic. Cyber-physical systems: the next computing revolution. In *Design Automation Conference*, pages 731–736. IEEE, 2010.